KAMAL GUPTA

San Francisco Bay Area, California, USA

email: kamalgupta308@gmail.com *web:* https://kampta.github.io

RESEARCH INTERESTS

My long-term goal is to build intelligent agents that can *see* (through vision, audio, and other senses), *interact* (navigate and act in an environment), and *reason* (plan long-term actions from sparse rewards).

EDUCATION

University of Maryland, College Park Aug. 2018 - May 2023
Ph.D. in Computer Science (4.0/4.0), Kulkarni fellow, Dean's fellow, Outstanding GRA Award
Advisors: Larry Davis, Abhinav Shrivastava
Thesis: Learning and Composing Primitives for the Visual World

Indian Institute of Technology Delhi Aug. 2007 - Dec. 2012
Bachelors + Masters in Electrical Engineering (8.6/10.0)
Advisor: Sanjiv Singh (Carnegie Mellon University)
Thesis: Pose Estimation for a Micro-Aerial Vehicle in GPS-denied environments

WORK EXPERIENCE

Tesla Bot (Optimus) May 2023 –
Staff Research Scientist *Palo Alto, CA*

- Training VLA (Vision-Language-Action) models with Imitation Learning and Reinforcement Learning for end-to-end whole body robot control. These models enable Optimus to navigate unstructured environments and manipulate objects in order to perform unsafe or repetitive tasks
- Scaling datasets and scaling models to ensure generalization and high success rate across wide range of tasks
- Developing eval suites to measure success rate, reliability, and speed of the robots for real-world deployment

Google Research May 2022 – Aug. 2022
Research Intern (web) *Mountain View, CA (Remote)*

- Image editing and model alignment with few in-the-wild images (accepted to ICCV'23)

NVIDIA AI May 2021 – Aug. 2021
Research Intern *Santa Clara, CA (Remote)*

- Generating textured meshes with VQ-VAE style two-stage generative models handing various 3D constraints

Amazon AWS May 2019 – Aug. 2019
Research Intern (web) *Pasadena, CA*

- One of the first works to apply autoregressive generative transformer models (GPTs) to data from visual domains such as 3D objects, Wireframes, Documents, etc. (accepted to ICCV'21)

NetraDyne Mar. 2017 – Aug. 2018
Staff Research Engineer *Bengaluru, India*
Netradyne provides ADAS devices for commercial vehicles. Led a team of 4 to ship

- Distracted (texting, eating, drinking etc.) + drowsy driving detection pipeline for driver safety
- DriverI Fleet Safety and Coaching Platform to ingest and analyze millions of driving miles

Poolka Technologies Apr. 2016 – Feb. 2017
Cofounder, CTO *Bengaluru, India*

- Built Fairi, a fashion assistant chatbot that provides clothing recommendations based on social media trends and users' existing wardrobe. (Pose estimation, Clothing segmentation, Graph Convolutions, Language Models)

Big Data Labs, American Express July 2013 – Mar. 2016
Research Engineer, Risk & Information Management *Bengaluru, India*

- Large scale recommendation systems, Geometric deep learning on financial data.
- **Platinum Genius Medal** - Systems and methods for customized real time data delivery Dec. 2015
- **Trainer of the Quarter** - Hadoop course for >200 AmEx employees in NYC, Gurgaon, Bangalore Dec. 2014

Robotics Institute, Carnegie Mellon University June 2011 – June 2012
Research Intern *Pittsburgh, PA*

- Developed an approach to predict vineyard yields automatically and non-destructively with cameras (web)
- Estimated global pose of MAV using stereo visual odometry fused with infrequent GPS measurements (web)

SELECTED PUBLICATIONS AND PATENTS

Visit Google Scholar tC3td8cAAAAJ for the complete list (⋆ denotes equal contribution)

- Measuring Style Similarity in Diffusion Models *ECCV* 2024 [web, code]
  G. Somepalli, A. Gupta, **K. Gupta**, S. Palta, M. Goldblum, J. Geiping, A. Shrivastava, T. Goldstein
- EAGLES: Efficient Accelerated 3D Gaussians with Lightweight EncodingS *ECCV* 2024 [web, code]
  S. Girish, **K. Gupta**, A. Shrivastava
- LEIA: Latent View-invariant Embeddings for Implicit 3D Articulation *ECCV* 2024 [web, code]
  A. Swaminathan, A. Gupta, **K. Gupta**, SR Maiya, V. Agarwal, A. Shrivastava
- LiFT: A Surprisingly Simple Lightweight Feature Transform for Dense ViT Descriptors *ECCV* 2024 [web, code]
  S. Suri⋆, M. Walmer⋆, **K. Gupta**, A. Shrivastava
- SHACIRA: Scalable HAsh-grid Compression for Implicit Neural Representations *ICCV* 2023 [web, code]
  S. Girish, A. Shrivastava, **K. Gupta**
- Chop & Learn: Recognizing and Generating Object-State Compositions *ICCV* 2023 [web, code]
  N. Saini, H. Wang, A. Swaminathan, V. Jayasundara, B. He, **K. Gupta**, A. Shrivastava
- ASIC: Aligning Sparse in-the-wild Image Collections *ICCV* 2023 [web, code]
  K. Gupta, V. Jampani, C. Esteves, A. Shrivastava, A. Makadia, N. Snavely, A. Kar
- Teaching Matters: Investigating the Role of Supervision in Vision Transformers *CVPR* 2023 [web, code]
  M. Walmer⋆, S. Suri⋆, **K. Gupta**, A. Shrivastava
- LilNetX: Lightweight Networks with EXtreme Compression & Structured Sparsification *ICLR* 2023 [web, code]
  S. Girish, **K. Gupta**, S. Singh, A. Shrivastava
- Neural Space-Filling Curves *ECCV* 2022 [web, code]
  H. Wang, **K. Gupta**, L. Davis, A. Shrivastava
- PatchGame: Learning to Signal Mid-level Patches in Referential Games *NeurIPS* 2021 [web, code]
  K. Gupta, G. Somepalli, Anubhav, V. Jayasundara, M. Zwicker, A. Shrivastava
- LayoutTransformer: Layout Generation and Completion with Self-attention *ICCV* 2021 [web, code]
  K. Gupta, J. Lazarow, A. Achille, L. Davis, V. Mahadevan, A. Shrivastava
- The Lottery Ticket Hypothesis for Object Recognition *CVPR* 2021 [web, code]
  S. Girish⋆, S. Maiya⋆, **K. Gupta**, H. Chen, L. Davis, A. Shrivastava
- Improved Modeling of 3D Shapes with Multi-view Depth Maps *3DV* 2020 [web, code]
  K. Gupta⋆, S. Reddy⋆, K. Shah⋆, A. Shrivastava, M. Zwicker
- PatchVAE: Learning Local Latent Codes for Recognition *CVPR* 2020 [web, code]
  K. Gupta, S. Singh, A. Shrivastava
- Applying Multi-Dimensional Variables to Determine Fraud USPTO 16/426826, 2019
  K. Gupta, V. Jain
- Systems and methods for Updating Fraud Detection Variable USPTO 15/258880, 2018
  K. Gupta, V. Jain
- Systems and methods for customized real time data delivery USPTO 14/961614, 2015
  S. Sanyal, S. Purkayastha, T. Choudhuri, A. Choudhary, V. Grover, M. Naeem, P. Mehta, **K.Gupta**, A. Agarwal

VOLUNTEER INITIATIVES

- **Graduate Student Mentor**: CVPR Academy for first-time CVPR attendees 2022
- **Co-organizer:** SIGGRAPH RCDC mentorship program for Graphics graduate school applicants 2021
- **Instructor**: AI4ALL summer program for high school students 2020
- **Reviewer**: Computer Graphics - SIGGRAPH 2022, SIGGRAPH Asia 2023. Computer Vision - ECCV 2022. CVPR 2023, 2022⋆, 2021. ICCV 2021. Machine Learning - NeurIPS 2022, 2023. ICLR 2022. (⋆ denotes Outstanding Reviewer Award)

MISCELLANEOUS

- Kulkarni Research Fellowship (web) 2020
- Dean's fellowship for outstanding academic achievement 2018-20
- **Coordinator**: Robotics Club IIT Delhi 2008-10
- Master's Research Scholarship by Ministry of Human Resources and Development, Govt. of India May 2011
- CBSE Merit Scholarship for All India Rank 38 (600,000 appeared) in AIEEE, now called JEE-Main 2007
- National top 1% (40,000 appeared) in Physics Olympiad and Chemistry Olympiad 2006
- National Talent Search Exam Scholarship by Govt. of India 2005